# Rajan Patkar

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## Highlights:

- Master's student at the University of Illinois seeking a full-time job starting 2027/05
- Researching cybersecurity for critical infrastructure & cyberphysical systems
- Hold current Secret-level clearance, Scholarship for Service CyberCorps recipient

## Experience:

Intern in Energy Systems at MIT Lincoln Laboratory (2026/05 - 2026/08):

- Owned project on high-fidelity simulations of operational technology (OT) networks
- Investigated disparate documentation to construct comprehensive NetBox OT inventory
- Devised pipeline from NetBox data to interactive GNS3 OT network simulation
- Demonstrated utility of simulation and proposed rehosting OT firmware on QEMU VMs

Research Assistant (prev. Intern) at the University of Illinois (2026/08 - now, 2025/05 - 2026/05):

- Targeting improving security and reliability within OT networks
- Identified and mitigated publicly known vulnerabilities on network-enabled OT devices
- Examined DPU performance in monitoring kernel health of networked devices
- Evaluated implementation of CIE practices for mitigating OT attacks

Cybersecurity Intern at the University of Illinois Information Trust Institute (2024/05 - 2024/08):

- Crafted hardware/software suite to determine efficiency of large-scale SSL offloading
- Designed and deployed network agents (C, Python) to record performance data
- Presented findings and recommended OT networks continue with on-device SSL

Engineering Intern/Student Trainee at Caterpillar (2022/05 - 2022/08, 2021/06 - 2021/08):

- Computerized collection and processing of machine data reports with Python
- Identified serial data communication problems (CAN, J1939) on engine control modules
- Automated testing of hydraulic excavator calibration software with MATLAB

## Skills:

- Proficient in C, C++, x86/RISC assembly, Python, MATLAB, R, Java, JS, and ArcGIS
- Adept communicator (native English/Marathi, intermediate Spanish/Korean)

## Education:

M.S. in Electrical and Computer Engineering at the University of Illinois Urbana-Champaign

- Prospective graduation and thesis submission 2027/05, curr. 3.69 GPA

B.S. in Computer Engineering at the University of Illinois Urbana-Champaign

- Graduated 2025/05, 3.72 GPA (incl. 2024 fall sem. at Hanyang University, Seoul, SK)